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SUMMARY

Results-driven Integrated M.Tech Computer Science Engineering student and patent holder specializing in machine learning pipelines, distributed systems, and backend engineering. Demonstrated ability to build production-grade systems achieving measurable outcomes - including a patented indoor positioning solution with $R^2 = 0.978$ accuracy. Experienced across the full stack from edge hardware integration to LLM-powered SaaS platforms.

EXPERIENCE

Patent: Coordinated Indoor Position Determination via Multi-Stage Signal Processing *Dec 2024–Dec 2025*

VIT, Vellore, IN

Application No. 202541115892

- Developed an end-to-end indoor localization system using Wi-Fi RSSI fingerprinting and Bluetooth integration, achieving approx. **1.6 m positioning accuracy (MAE: 1.468 m, R^2 : 0.978)** in complex indoor environments.
- Built an ESP32-based autonomous Wi-Fi sniffer mounted on a robotic car, automating radio-map generation and reducing manual site-survey effort while collecting **4,277 RSSI records across 209 access points**.
- Designed and optimized a Dynamic Stacking ensemble model combining KNN, Random Forest, and XGBoost with a Gradient Boosting meta-learner, improving localization performance by **approx. 43% (RMSE: 3.124 to 1.787)** over conventional stacking approaches.
- Engineered a real-time data pipeline using MQTT, Raspberry Pi, and Python, transforming raw Wi-Fi scans into machine-learning-ready datasets and enabling location predictions within the system's real-time constraints.
- Implemented an **A*-based indoor navigation** module using NetworkX, generating optimal routes from predicted user locations to destinations and enabling dynamic path recalculation for real-time indoor guidance.

PROJECTS

AutoML Assistant

Jan 2026

Project

Repo | Live Demo

- Built an end-to-end AutoML platform using Streamlit + Python to automate data upload, preprocessing, analysis, model training, and optimization workflows.
- Developed a modular ML backend supporting **regression, classification, and time-series forecasting** with algorithms like XGBoost, Random Forest, ARIMA/SARIMAX, and optional LSTM.
- Integrated **LLM-powered model recommendation (RAG)** using LangChain, Groq (Llama 3.1), and FAISS to suggest suitable models based on dataset statistics and ML best practices.
- Implemented **SHAP-based explainability** (feature importance, beeswarm, waterfall, dependence plots) to make model predictions interpretable for end users.
- Exposed trained models through a FastAPI inference API (`/predict`, `/predict/csv`) with **persistent workspace/model management** for reusable deployment-ready ML pipelines.

OncoFollow - Full-Stack Oncology Follow-Up Platform

Apr 2026 – Jun 2026

Project

Repo | Live Demo

- Built OncoFollow, a full-stack clinical workflow platform in **Next.js 14 and React 18** that unified patient symptom reporting, clinician alert queues, admin oversight, and longitudinal care timelines - reducing manual triage overhead and improving clinician response time to high-acuity cases.
- Designed and implemented a **serverless API layer** (Next.js API routes) proxying Groq Cloud LLM inference with role-specific prompt controls, server-side secret management, and structured JSON contracts - applying production-grade **cloud architecture** and **API design** patterns.
- Engineered a rule-based triage engine spanning **20 symptom categories** across 4 clinical scoring dimensions (severity, duration, frequency, trend), with a safety-first escalation layer featuring red-flag detection, symptom-combination overrides, and output sanitization - achieving **99% sensitivity** for emergency-case interception in synthetic evaluation.
- Developed real-time **analytics dashboards** with interactive data visualisation (Recharts), including lab-trend sparklines, NEWS2 acuity scoring, and AI-powered trend analysis - mirroring the sensor-to-insight pipeline used in IoT-driven clinical environments.

- Grounded all AI recommendations in NCCN/ASCO clinical guidelines and patient records, producing citation-backed outputs ([R1], [G1]) for clinical traceability - demonstrating awareness of **HIPAA-adjacent compliance** requirements including audit logging, consent management, and blocked-pharse filtering.

FLEX-DCA AI Platform - AI-Powered Debt Collection Allocation & Compliance System
Jan-Feb 2026

Project
Repo

- Built a full-stack debt recovery platform using FastAPI, React, and SQLite to ingest cases, run predictions, auto-allocate DCAs, and track operational KPIs via REST APIs.
- Developed and integrated **3 CatBoost models** (recovery probability, aging risk, closure speed) with financial feature engineering for real-time decision support.
- Implemented a RAG-based legal copilot using FAISS + SentenceTransformers + Groq LLaMA 3.1, enabling policy-grounded answers for FDCPA, Regulation F, and RBI compliance queries.
- Created an operations dashboard with KPI monitoring and case workflows, improving visibility into recovered/open cases and allocation outcomes.
- Structured backend into modular services (ML, allocation, RAG, dashboard, ingestion) to support scalable, production-style architecture for enterprise debt operations.

EDUCATION

Vellore Institute of Technology , Vellore	<i>Expected April 2027</i>
Integrated M.Tech in Computer Science Engineering	GPA: 7.0/10
<i>Relevant Coursework: Machine Learning, Distributed Systems, Data Structures & Algorithms, Database Management, Computer Networks, Operating Systems, Cloud Computing</i>	
SM Public School , Haridwar	<i>2020–21</i>
Junior College (Class XII)	91.2%
SM Public School , Haridwar	<i>2018–19</i>
High School (Class X)	87%

TECHNICAL SKILLS

Languages & Core: Python, C++, Java, JavaScript, SQL
Backend & Systems: FastAPI, REST APIs, Microservices, MQTT, Docker, CI/CD
Databases: PostgreSQL, SQLite Tools & Cloud: Git, AWS, GCP, Streamlit
ML & AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, LSTM, SHAP, RAG, FAISS, LangChain, Groq
Methodologies: Agile, Scrum, DevOps, Data Engineering, Big Data

LEADERSHIP & EXTRACURRICULARS

Sponsorship Coordinator , ISTE Chapter – VIT Vellore
• Secured Rs. 1,00,000+ in corporate sponsorships for flagship hackathon (Horizon) by pitching to 100+ founders and startups; orchestrated end-to-end event logistics and developed outreach frameworks that established strategic industry connections for future initiatives.

ACHIEVEMENTS

1st Place, ARC Hackathon (ARC Lab, VIT) – Built a GUI-based AutoML tool that automatically benchmarks and recommends the best ML model for any given dataset, winning against competing teams.

CERTIFICATIONS

Stock Market Basics & Technical Analysis – View Certificate
Designing Seamless Notification Service – View Certificate